Research on the Townscape Management Strategy of Small Towns in Urban Suburbs—Take Dujiangyan City as An Example

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Abstract: ***Since the reform and opening up, China's urbanization is advancing rapidly, promoting the majority of small towns to carry out the renewal construction in full swing. But there are some small towns in the process of construction ignoring their own actual consideration. To unilaterally imitate the experience and solutions of other small towns, making the style of small towns in different regions gradually become similar, street landscape, spatial pattern, architectural image gradually presents the phenomenon of homogenization, and there is an increasingly severe trend. As a city in northwest in Chengdu, the small towns surround in Dujiangyan are very typical in China. This article based on the townscape analysis of small towns in Dujiangyan, putting up several questions and relevant optimized measures, which wish to give similar urban development and style construction to make a certain reference.***

Keywords: Townscape; Small town; Smart management; Urban planning.

1. RESEARCH BACKGROUD

Since the reform and opening up, China's urbanization is advancing rapidly, promoting the majority of small towns to carry out the renewal construction in full swing. All of a sudden, the face of small towns has changed dramatically, n this process, as the system and experience is still in the stage of "crossing the river by feeling for the stones", People are still relatively backward in their thinking, In addition, the great contradiction between the rapid modernization construction and the resources, culture and environment of small towns themselves, There are some small towns in the process of construction to ignore their own actual situation consideration. To unilaterally imitate the experience and solutions of other small towns, making the style of small towns in various regions gradually become similar, street landscape, spatial pattern, architectural image gradually presents the phenomenon of homogenization, and there is an increasingly serious trend[1,2].

Since the 18th National Congress of the Communist Party of China, China's social and economic development model has formally put forward the transformation, and the new urbanization strategy has been put forward. "Establishing small towns with distinctive characteristics and full of charm" has become the central idea of the construction of small towns in the new era. Small town style is a requisite part of town culture. The study of townscape can help people to better understand the local history and culture. Additionally, the urban style requires the consideration of the urban humanistic elements[3], the protection and inheritance of the townscape of small towns can help to protect the cultural heritage of the towns, so that future generations can better understand and inherit the culture. Small towns are constantly facing the contradiction between development and protection. In the context of globalization, and the process of understanding urban style, measures should be taken according to local conditions to reduce randomness and increase locality[4]. Intelligent management mode should be applied to urban management. The intelligent management of communication, circulation industry, gas and other infrastructure can realize the early warning and detection on the official website, improve the quality of community, improve the degree of information governance, and promote the development of industrial environment system in different regions[5,6,7].The study of the style of small towns is helpful to improve the cognition of urban and rural development, and promote the urban and rural integration and sustainable development. Many small towns have their unique architecture, scenery and historical culture, so they have high tourist attraction. Studying the features of small towns can better understand the tourism resources and provide help for the development and promotion of tourism[8]. Studying the townscape of small towns can improve the level and quality of town planning and create a better urban environment, can also provide important support for the local economic development. Townscape renovation can make the buildings scattered, the perfect combination of ancient rhyme and modern, integrated presentation, to help realize the revitalization of the village[9]. While protecting and inheriting local cultural heritage, resources can be used to carry out economic activities, such as cultural and creative industries, rural tourism, etc., To promote local economic development and improvement of people's livelihood. Therefore, it is of great significance to research the strategies of the townscape improvement.

Dujiangyan city with mountains, water, forests, weir, bridges integrated, which is a natural ecological garden city in the northwest of Chengdu, Known as the city layout characteristics called "irrigation city water color half city mountain". The suburban towns of Dujiangyan city also have a single traditional townscape and characteristics called "a thousand towns, one face". Therefore, through a thorough analysis of the landscape status of small towns in the suburbs of Dujiangyan city, summarize the problems existing in the current situation. Based on the small towns in the inner suburbs of Dujiangyan City,

classification of the improving strategies of small towns in urban suburbs, which can provide a reference for a kind of small towns with the same characteristics in the style renovation. It will expand the case reference for the study of small town style renovation and promotion strategy.

2. DUJIANGYAN CITY BASIC SITUATION OVERVIEW

2.1 Dujiangyan city basic situation overview

Dujiangyan City is located in the northwest corner of Chengdu city, formerly known as Guan County. In May 1988, it was removed and established as a county-level Dujiangyan City. By November 2020, it had a population of 710,100 and a total area of 1,208 square kilometers.

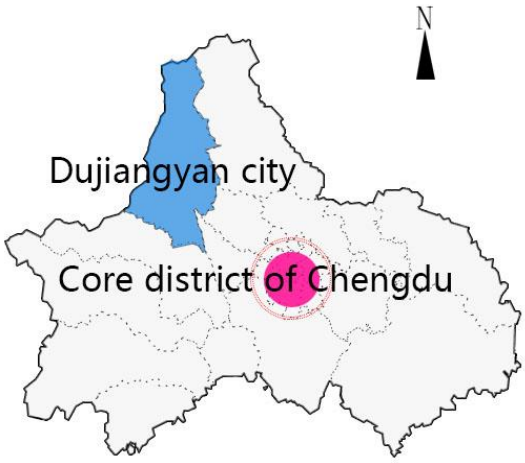


Figure 2-1: schematic diagram of Dujiangyan City in Chengdu City (source: Self-drawing)

2.2 Overview of the characteristics of Dujiangyan towns

As of October 2021, Dujiangyan city has six streets and five organized towns under its jurisdiction, among which five organized towns is Longchi Town, Qingchengshan Town, Juyuan Town, Tianma Town and Shiyang Town, which are located in the mountainous area to the north of the urban area, the shallow hill in the south of the urban area and the flat area in the southeast of the urban area, as shown in Figure 2-2.



Figure 2-2: administrative division Map of Dujiangyan City (source: modified by the author)

Longchi Town is located in the northern mountainous area of the Chengdu city, the closest to the Dujiangyan urban area, and is listed as an important water source of Dujiangyan City and even Chengdu City. There is Longchi National Forest Park Reserve in the town, which is less affected by the urban development, and the town is dominated by natural landscape features. Qingchengshan Town is located in the southern area shallow hill area, "the world natural and cultural heritage, Chinese Taoism birthplace" —— Qingcheng mountain scenic spot in Qingchengshan Town, so the Qingchengshan Town has distinctive natural landscape scenery and profound history and culture, whose economic development is given priority to with tourism, and a large number of tourists to visit every year. Influenced by urban radiation, the townscape is given priority to with natural ecological landscape and historical cultural landscape. Juyuan Town is located in the southeast of the city. It is an important transportation station from Chengdu to Dujiangyan and even Aba Prefecture, which makes Juyuan Town obviously driven by the city, and the town is mainly ordinary townscape, with few obvious prominent characteristics, and even slightly chaotic. Tianma Town is located in the southeast of the city, at the junction of Pidu District, Pengzhou City and Dujiangyan City. It is a largely agricultural town dominated by agricultural industry and rich types of agricultural products, so the town is mainly agricultural landscape. Shiyang Town is located in the south of the city, is another agricultural town under the jurisdiction of Dujiangyan City, has a long history. As early as the Han Dynasty, is a famous town, is the hometown of Lady Huarui, or the site of ancient Qingcheng County, has a certain historical and cultural accumulation.

Table 2-1: classification of small towns

No.	Name of town	Type classification	Classification of geomorphology	Other attributes of small towns
1	Longchi Town	Ecological protection of small town	Small towns in mountainous areas	Ecological protection areas, prohibited development zones, environment. There is a rural national forest park
2	Qingchengshan Town	Characteristic tourism-type small town	Small towns in hilly areas	National key town, Chengdu key town
3	Juyuan Town	Industrial and commercial small town	Dam area small town	Dujiangyan city, "East river important town"
4	Tima Town	Agricultural small town	Dam area small town	General small towns
5	Shiyang Town	Agricultural small town	Dam area big town	Provincial rural culture revitalization model town

2.3 Dujiangyan suburban small town landscape status analysis and summary

Through the investigation of the small towns, it can be found that the townscape of the small towns has their own characteristics, but there are some common problems, such as the townscape is a certainly similarity. It has a typical common characteristic of small towns, therefore, we can four cases —Longchi Town, Qingchengshan Town, Juyuan Town, Tima Town— of small towns as a whole, to obviously prominent, influential larger style problem is summarized.

2.3.1 Natural ecological environment

2.3.1.1 The rich river system resources are not fully utilized

Dujiangyan city is a city near mountains and rivers. It is famous for its rich water resources and rich plant resources and water system resources, creating this livable urban environment. Dujiangyan city water resources into the streets of the city, participate in shaping city street style, enrich the level of urban style. Nevertheless, the suburban small towns of river system resources utilization degree are not high, it mainly displays in the waterfront space construction level is not rich, architectural landscape is monotonous, participation of citizens in activity is not high. It ignores the function of the river shoreline of space for small towns near, eventually makes the small townscape lack water landscape participation, waterfront ecology, architecture, cultural landscape are relatively lack.



Figure 2-3: current situation of Longchi Town and surrounding areas (source: self-taken)



Figure 2-4: case status of river system in small towns (source: self-shot)

2.3.1.2 The geomorphic terrain changes greatly, Terrain utilization is not high

As the town itself is in the mountainous environment, the overall terrain actually has the elevation change, and the small towns in the western region are limited by less flat land and more slopes and mountains. In fact, such an environment can be built according to the mountain situation. Chongqing, Hong Kong, Chengde, Yangquan and other cities are all models of using the terrain, while the towns of Dujiangyan do not use the terrain to form key landmarks, nor did they form the corresponding terrain style. This phenomenon is also particularly prominent in other small towns in mountainous areas, which has great constraints on the future development of small towns in mountainous areas.



Figure 2-5: an elevation map of Dujiangyan City (left), a satellite map of Longchi Town Town (right) (source: self-shot, provided by the Internet map)

2.3.2 Living condition

2.3.2.1 Urban functional structure is unreasonable, and the architectural style is disorderly

The functional structure division of land use in small towns was originally relatively simple, mainly to the living function of daily life. So,it is mixed in the space is more, additionally, the management is not perfect， So that all kinds of land cross-use exist. The influence of this in townscape is mainly reflected in the poor continuity of urban block section,old street facade. Architecture is the most numerous and largest physical element in small towns, it is also the basic element of the small townscape. How to shape a unified, coordinated architectural style with certain regional characteristics has always been the most concerned, and the trickest problem. Based on this, through in-depth and detailed field research in the cases, it is concluded that the architectural style of small towns in Dujiangyan is hardly shaped in two aspects. The first is the mix of architectural styles, and the second is the mix of architectural colors.

The architectural style is the external image of the building. Due to the generally large construction time span of buildings in small towns, design or construction personnel to concern about different aesthetic elements in different periods, n addition, the construction activities of small towns have been lack of formal design or construction management, so it is difficult to form a relative unity. Modern type, antique, lift eaves, flat roof, all kinds of architecture styles are uneven. In the above small towns, the number of building is mainly low multi-storey,The roof form is mostly sloping roof. Although it seems that there will be a certain degree of unity on the surface, there are still such as modern style roof and European arch window, European column and other elements, or with the Chinese style elements such as the peak ridge, the Chinese eaves of the beast fairy. In addition, the lack of unified planning in the architectural design and construction of small towns caused by the great difference of window to wall ratio and other architectural details, making the case of small towns architectural style chaotic, as shown in Figure Figure 2-5.



Figure 2-5: current Status of Living Environment in Qingcheng Mountain Town (source: self-taken)

As for color, so the building color is the main color of urban material environment, reflected by the cover on the surface of material color properties. According to the composition of three-stage architecture, a building color can be divided by different roof colors, building outside metope color, other parts of the building, other parts of the color mainly attached to the building facade advertising signs. About roof colors, above small town is an example, be for need of water prevention and drainage, roof form is in the majority with a slope roof, use flat roof less. The color is mainly the typical gray roof of western Sichuan dwellings, but other colors are rare. This promotes the coordination of small towns to some extent, which is the common advantage of small towns in architectural color.

In addition to slope roof building, roof form for flat roof of small towns, for the top sun cooling, increase the drying area and increase the use area. Homeowners built on their flat roofs with blue, gray or yellow, red color steel shed, or building houses, for the reason that this kind of construction is lack of unified planning and management, combined with the added part color lightness is high, in the overall environment, to some extent, lower the overall environment of small towns` beauty, as shown in figure 2-6.



Figure 2-6: example of buildings built in different ages in Juyuan Town Town (source: self-photography)

Small towns` buildings exterior wall color mixed as the situation shown by the roof, this is because the different periods of different exterior wall decoration materials, building present the color saturation, lightness, color, such as effect difference. Small towns building flexibility is larger, the owner can choose exterior wall decoration materials according to their own needs. So it is difficult to form a relative unity, finally form the current small towns overall show large area of monotonous color exist more small area jumping patches.

Building facade advertising sign color is another important factor of building facade color, attached to the building facade advertising signs installed in more striking position. Businesses in order to attract people, tend to show the billboard with abrupt color, large size, make it appear more dazing, especially in the main road and important road node this phenomenon is particularly obvious, as shown in Figure 2-7.



Figure 2-7: case current situation of color element composition in streets and alleys in some towns (source: modified by the author)

2.3.2.2 Street contours lack a sense of hierarchy, and continuity is destroyed by too many additional elements

The street contour line is composed of the building, the surrounding mountains, the water body,the blue sky and the white clouds and other elements, which is a unique symbol highlighting the cultural, aesthetic and modeling characteristics of the city.The special shaping of street contour lines is more common in cities, especially the famous metropolises have already realized the importance of promoting the city image and highlighting the charm of the city.It was specially built early, making it a symbol of the city, forming a unique skyline landscape.

For small towns, the street outline is difficult to reflect its characteristics.The main reasons have two aspects: one is the backward management of the top buildings of too many buildings; the other is that the small buildings are low, mostly two or three stories, the number of floors is not fluctuating, the lack of rhythm changes, the horizontal line is rigid and uncharacteristic, as shown in Figures 2-8.



Figure 2-8: case schematic diagram of the skyline composition of some streets in small towns (source: modified by the author)

The auxiliary facilities of the top floor of the building refer to the water heater, the water tank, metal color steel shed, movable board houses and advertising signs added after the completion of the main body.These additional ancillary facilities are often sharp in color and difficult to integrate with the main body of the building. They exist abruptly in the sight corridor of the streets of small towns and become an important factor affecting the coordination of the skyline of small towns, as shown in Figure 2-9.



Figure 2-9: Case Example of various ancillary facilities added on the roof of a small town (source: self-photo)

Building number of small towns for six and six layers, more for two or three layers, and the cornice founder of modern style “square box” building, these buildings have similar appearance and scale, so the formation of the skyline is given priority to with horizontal line, turn crisp, overall plain lack of change, the lack of administrative levels.

2.3.2.3 Low level of infrastructure construction

Although the level of infrastructure construction in small towns is higher than that in rural areas, compared with the cities, the construction standard, the coverage rate and service quality are still far behind the cities, which brings great inconvenience to the lived experience of small town residents. Similar to most small towns in China, the problems of unbalanced infrastructure allocation and poor service effect in small towns are obvious.

Taking the small towns as an example, the deficiencies in infrastructure construction are mainly reflected in the following three aspects. First, there is a widespread problem of low coverage of municipal public service facilities such as street lamps, dustbins and public toilets. This problem directly leads to the reduction of the use of public space in small towns.Second, the damage to public facilities cannot be repaired quickly. This problem not only affects the convenience of using public space, but also affects the external image of small towns. Third, the cable line erection rate in the air is high, and it is urgent to make a large-scale "cable into the ground" transformation, as shown in 2-10. The erection of cable lines in the air will not only affect the overall environment of small towns due to the formation of messy line network in the air by private connection lines, but also bring safety risks due to the collapse of telephone poles and the falling off of lines. It may also be due to the collapse of telephone poles, lines fall off to bring safety risks.



Figure 2-10: case example of the common cable line erection phenomenon in small towns (source: self-photo)

2.3.3 Cultural Environment

Cultural landscape is an important part of the urban landscape, and the cultural environment also constitutes an important part of the urban landscape. Regional history and culture is the spiritual memory formed by the town in the long history. It is because a town can move forward and get better and better in the current of The Times because of its unique cultural heritage. Therefore, it is particularly important to fully excavate the regional cultural resources contained in the landscape renovation of small towns.

As a city with a long history, Dujiangyan city itself has a very high historical and cultural origin. The water conservancy project of Dujiangyan has not only solved the problem of watering the Tianfu Plain in Chengdu, but also become a famous scenic spot. Qingcheng Mountain is a national 5A scenic spot. While having beautiful scenery, sun Simiao, Du Guangting and other famous celebrities and cultural allusions are stored here, and there are more legends. Bai Suzhen cultivation, Ning Fengzi cultivation, Li Bing dragon all come from here.

However, in addition to the legend of Dujiangyan itself, each town also has unique historical and cultural resources, which it need to be inherited to create the unique features and characteristics of the town. According to the field investigation, from the perspective of historical cultural heritage, case of small towns generally have a certain degree of historical accumulation, but the case of small towns generally make full use of the history of human resources, mining, protection, inheritance, and make them face the risk of extinction, as shown in Table 2-2.

Table 2-2: regional historical and cultural resources and inheritance status of small towns

Town name	Historical resource name and its type	current status
Longchi town	Tianguan Festival —— Folk culture activities	Natural disasters lead to their loss and low interest among young people
Qingchengshan Town	Yao Wang Temple meeting ——folk culture activities	The audience is mostly for middle-aged and elderly people, and it is difficult to attract the interest of young people
Juyuan town	Juyuan bamboo carving —— provincial intangible cultural heritage	The old artist dies, but the younger generation is not interested and faces a loss
Tianma town	Tianma sedan suona —— county level intangible cultural heritage	Family inheritance leads to a small spread and the risk of losing skills

3. DUJIANGYAN CITY STYLE DEVELOPMENT REQUIREMENTS AND STRATEGES

Obviously, there are still a lot of problems in the urban landscape structure of Dujiangyan city, which directly interferes with the operation of the urban landscape. Therefore, the urban landscape structure of Dujiangyan city must be gradually optimized and gradually improved according to the positioning of Chengdu city. According to the *Territorial Space Planning of Chengdu (2020-2035)*, Chengdu plays an important role as an economic, technological and financial center in the country.In the field of cultural construction, it is necessary to focus on building "three cities and three capitals", namely, the world tourism city, the world cultural and creative city, the world competition city, the international food city, the international music city, and the international exhibition capital.The development requirements of Dujiangyan city echo the requirements of Chengdu, and the goal is to build a world tourism destination, through building the world heritage landscape of peace and Yicheng, and to achieve this goal.

It can be said that the planning not only implements the current situation, but also meets the requirements of the future construction of Dujiangyan in Chengdu. Therefore, the small towns of Dujiangyan city should reasonably improve their townscape under the condition of meeting their own basic conditions, and create a unique townscape that meets the requirements of Chengdu and Dujiangyan. According to the superior planning requirements and the preliminary results of urban research, the following basic requirements for an style control and construction are put forward.

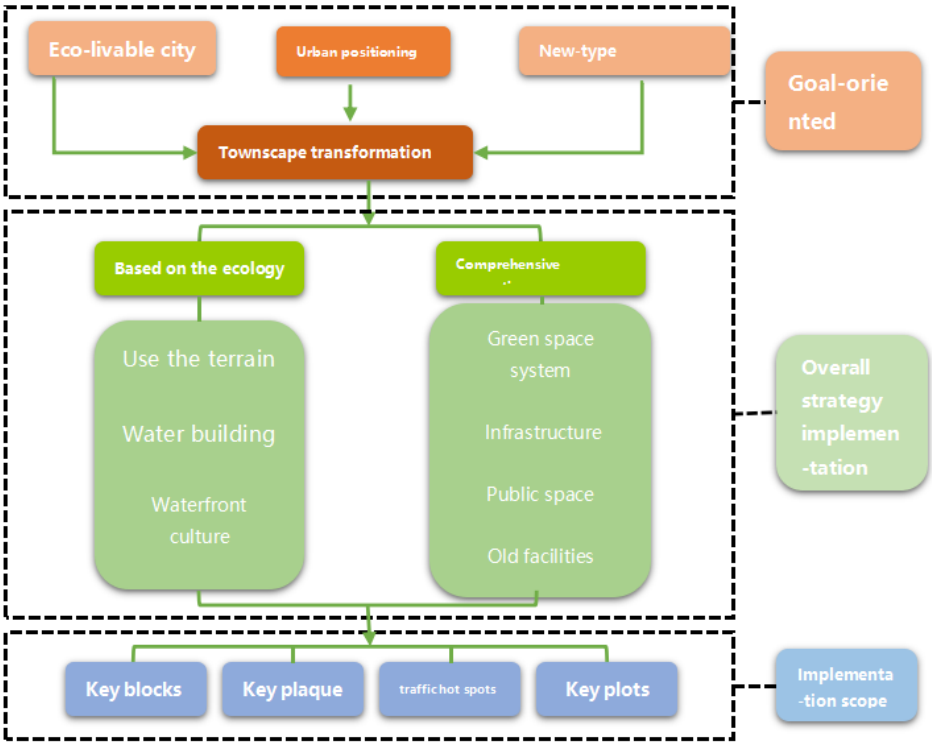


Figure 3-1: Case Example of cable line erection in small towns (source: modified by the author)

3.1 Put the people first, respect the nature, and establish the protection elements

The core concept of urban and rural planning is to put people first and respect nature. In fact, the reconstruction of cities and towns is not equivalent to the demolition and reconstruction, which may lead to the increase of the government's financial investment and bring an unknown burden to the urban residents. For Dujiangyan city, its natural and cultural conditions are superior, so it is more necessary to emphasize the protection and rational utilization of the natural environment, and to shape the landscape and style with the help of the terrain. Human-centered planning requires a systematic investigation, which will reflect the development status and future direction of Dujiangyan City. Respecting nature does not mean refusing to develop, otherwise the value of urban and rural planning will be weakened. Dujiangyan city needs to strike a balance between development and nature protection, and to seek development space on the basis of protection. After scientific investigation, protection elements should be prioritized. Next, Dujiangyan city needs to formulate detailed protection measures, and gradually promote the implementation according to the actual situation. In this process, the needs of urban residents should be fully considered, and the excessive exploitation of natural resources should be minimized to avoid environmental deterioration. In the process of development, the planning should be adjusted timely, and the urban and rural planning scheme should be continuously optimized to better meet the actual development needs of Dujiangyan city.

3.2 Based on the international community, display the culture, and build the world landscape

According to the development strategy of “three cities and three capitals” advocated by Chengdu municipal government, we need to fully realize that in the planning and construction of small towns, we must base ourselves on the objective development conditions of their region, and at the same time draw lessons from the world's advanced international vision, so as to show the cultural heritage of the town. For the many small towns in Dujiangyan area, we should not only fully explore their historical and cultural connotation, but also effectively integrate modern technology to fully display the beauty of mountains and rivers. In the previous chapters, we mentioned that these towns had their own unique cultural solar term activities, which should have been retained and promoted, but also in line with the development needs of the world heritage landscape Anyicheng. In addition, we should also encourage the small towns to give full play to their own unique geographical advantages, and on this basis, to further expand the tourism industry related to the well-known culture at home and abroad. On the other hand, the abundant water resources and unique geographical advantages of Dujiangyan area provide unique conditions for agricultural development. According to the requirements of the agricultural industry, it is also possible to appropriately build a green agricultural characteristic town. In this way, these towns can not only fully display their own characteristics and make contributions to the local economic and social development, but also add luster to the world heritage landscape in Dujiangyan area.

3.3 Improve the foundation, optimize the facilities, and create a livable life

In the basic research, it has been found that the infrastructure construction of the small towns around Dujiangyan is not perfect, which not only affects the basic use function of the town, but also has a bad impact on the style of the town. These messy electric piles, messy garbage bins, and dilapidated public services are like many traditional towns that need to be improved. This is extremely inconsistent with the positioning and function of Dujiangyan City, which not only affects the living environment of urban residents, but also greatly affects the impression and experience of tourists. We must take action that to optimize the infrastructure of small towns, improve the quality of infrastructure, and then improve the quality of people's life. First, a comprehensive analysis and evaluation of the infrastructure to find out the existing problems and deficiencies. These problems may involve urban roads, Bridges, hydropower stations, garbage treatment facilities and other aspects. We need to fully understand the current situation, find out the existing problems, and then take targeted measures to solve them. Secondly, a scientific and reasonable action plan should be formulated, and the construction and improvement of infrastructure should be incorporated into the overall planning of urban development. The planning scheme needs to comprehensively consider the local natural environment, economic development level, the population structure, urban function and other factors, so as to ensure the scientific nature and feasibility of the planning scheme. Thirdly, the management and maintenance of infrastructure should be strengthened. The quality and functionality of the infrastructure depend on the level of management and maintenance. We need to establish a set of perfect management system, develop a scientific maintenance plan, to ensure the normal operation and good use of the infrastructure. Finally, publicity and education should be strengthened to improve residents' awareness of infrastructure protection.

3.4 Digital empowerment, intelligent supervision, and build a healthy city

In fact, Chengdu has entered a stage of highly intelligent development, including small towns around Dujiangyan. Digital governance and integrated application scenarios can not only be used in urban construction, but also be necessary in the construction of urban landscape construction. Such a high level of wisdom makes the whole Chengdu become a cashless metropolis. All daily consumption can be easily completed by scanning the code by mobile phone, or even swiping the face. Shared bikes and self-service vending machines, allowing citizens to enjoy the convenience and fun of the digital age. In Dujiangyan, a beautiful town, people feel a life of nature and modern technology. The pace of life here is comfortable without rhythmic. Walking in the streets of the town, you will be attracted by the modern and classical, nature and humanity here. This highly intelligent development should make the construction of urban landscape more accurate and in place. However, due to various reasons such as the capital epidemic, the landscape of the town has not been improved and implemented. Therefore, in the further construction, we should think of the digital empowerment mode of sustainable development, scientifically supervise the urban development, and make use of the construction opportunity of music capital and cultural and creative city to create a perfect digital Dujiangyan, and create a healthy urban small town circle.

3.5 Increase cooperation, innovate models, and find ways of implementation

In fact, China and many other countries have just emerged from the storm of the epidemic, and the economy is still recovering again, and the reconstruction of the old city itself is a big urban problem. Therefore, it is difficult to solve many problems, such as capital, land, residents' different opinions, and difficult government decisions. Therefore, the above problems cannot be separated from the cooperation of the government, residents, experts, universities, enterprises and other parties, only in this way to discuss and plan together, can effectively solve the problem. Because, need to have a new good cooperation or a research mode,the skills planning to think about the city style construction, and can effectively solve the capital problems, at the same time can be in line with the interests of the masses to solve the core problems of the masses, this can put the solution of the style problem into practice.

4. OPTIMIZATION MEASURES OF SMALL TOWNS IN THE SUBURBS OF DUJIANGYAN CITY

4.1 Optimization ideas

According to the development requirements and strategies of Dujiangyan city style, the specific key points and rough contents that need to transform the style can be clearly defined. However, in order to effectively solve the problem of townscape, it is necessary to have practical outcome measures. The implementation of these measures must take into account the living needs and requirements of the residents, the government and other relevant people in the cities and towns.Therefore, while considering the implementation of the above requirements and measures, questionnaires and interviews should fully collect the opinions of residents and relevant personnel, organize relevant meetings, and conduct face-to-face consultation and communication. Then, the unified continuous compilation, and formulate the relevant measures.

4.2 Optimization measures for small townscape

4.2.1 Implement overall planning, formulate regional planning, and carry out natural protection and improvement

Under the condition that the territorial space planning of Chengdu and Dujiangyan has been completed, in order to realize the comprehensive coordination and planning construction of small towns and strengthen the utilization and protection of the natural space environment, it is necessary to formulate the planning program of each town, that is, the territorial space planning at the town level. It is reported that at the beginning of 2023, the core area of Qingcheng Town, Longchi Town, Tianma Town and Juyuan Town have started the work of territorial space planning at the town level, marking the road of coordination of each area has been opened. This can be considered as an important optimization measure to implement landscape protection. However, the implementation of the plan is still the key to ensure the perfection of the plan. Therefore, the following planning scheme, further planning and design, and action program are all important parts of the planning implementation, and they are also part of the planning coordination. In order to realize a scientific planning organization, we need to gather opinions from all parties, form a planning program, and incorporate it into the protection and utilization of the natural space environment, so as to improve the urban space environment.



Figure 4-1: the core area of the four towns has been started and plan

4.2.2 Optimize the natural landscape and use the water system space to create a rich waterfront life

In small towns, it is necessary to improve the application of water system space and build a comprehensive water system landscape system. At the same time, we need to optimize the urban public space to enhance the participation of residents and enhance the interaction between people, so as to realize the harmonious coexistence between human beings and nature. We should allow the water system, plants, green space and other elements to create more opportunities for residents to participate and interact. According to the conditions, we can establish waterfront parks to provide places for leisure and entertainment, and small towns should plan and build waterfront parks around the water system, or naturally integrate the water system space, to ensure the efficient use of water resources and provide leisure and entertainment space for residents. In the park, water-friendly platforms, plank roads, seats and other facilities can be set up, so that residents can have close contact with the charm of water bodies. Strengthening the construction of waterfront walking and bicycle lanes to encourage residents to participate in outdoor activities, small towns should strengthen the construction of waterfront walking and bicycle lanes to ensure their connectivity and comfort. Walking paths and bicycle paths can be laid with environmentally friendly materials to reduce pollution to water bodies. To strengthen water ecological restoration and restore the self-purification capacity of water bodies, small towns should pay attention to the ecological restoration of water bodies, purify water quality and protect the water ecosystem through ecological floating islands and constructed wetlands. In addition, the supervision of water bodies should be strengthened to prevent pollutants from entering water bodies and ensure the cleanliness and self-purification capacity of water bodies. An

intelligent management system should be established to monitor water quality and ecology in real time. In order to ensure the sustainable utilization of water system space, small towns should introduce an intelligent management system to implement real-time monitoring of water body, water quality and ecosystem. The system should not only be able to provide real-time data of water body and water quality, but also timely report abnormal situations when found, so as to provide reference for subsequent treatment. The introduction of this intelligent management system can improve the efficiency of water quality management, reduce the risk of pollution, and provide residents with a safer and more comfortable water system environment.

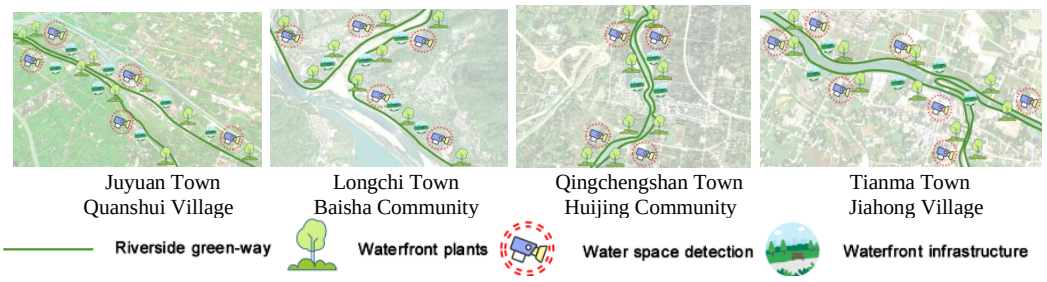


Figure 4-2: intention map of riverside space reconstruction in surrounding small towns (source: modified by the author)

4.2.3 Shape the characteristics of the town, carry out detailed design, and transform the overall appearance of the town

Systematic reform of the street style, the original trivial and chaotic street facade transformation. First of all, comb the main urban colors, main and secondary buildings, materials, summarize and classify these elements organically combined together to form a unique town style. Still the historical and cultural background of each town, exploring its unique regional characteristics, and integrate these characteristics into the street style design. In the design process, to ensure that the town style has both traditional charm, and modern atmosphere, with the town color, main material and core decorative elements constitutes the main tone of the town style. Secondly, we should strengthen the optimization of spatial layout, emphasize the construction of portal space, urban spatial nodes and urban landmarks, and strengthen the construction of key space of street and lane style. Thirdly, optimize the detail design, and make fine design in the street planning, the building facade, landscape greening and other aspects, to ensure that every detail is in line with the characteristics of the town. For example, in the building facade design, the local unique architectural elements, such as masonry, tiles, can be used to make the architectural appearance more distinctive; in terms of landscape greening, the local unique plants can be selected to increase the natural atmosphere of the town. Finally, strengthen the publicity and education of urban residents, improve the residents' understanding of the transformation of street style. By holding publicity activities and distributing publicity materials, residents can understand the importance of the street and lane style transformation, encourage residents to actively participate in the street and lane style transformation, and take the initiative to protect the urban style.



Figure 4-3: Reconstruction diagram of architectural elements(source: modified by the author)

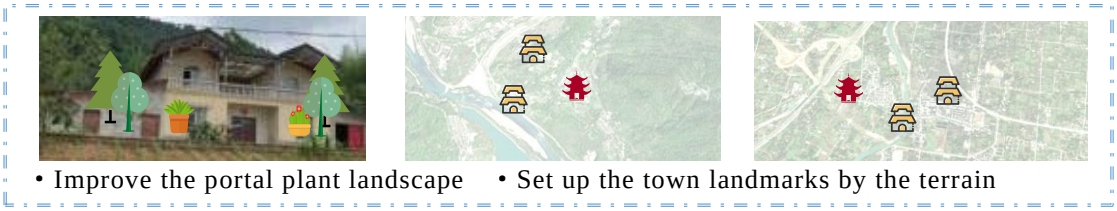


Figure 4-4: Reconstruction diagram of town space(source: modified by the author)

4.3.4 Improve the infrastructure, upgrade the environment, and improve the quality of life of residents

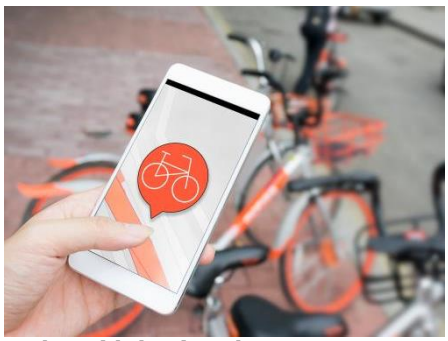
In order to further improve the quality of life of residents, the government and relevant departments need to constantly improve the infrastructure and upgrade the environment. First, we need to increase investment in infrastructure and improve environmental, transportation, communications, energy and other infrastructure. We will strengthen the upgrading of urban roads, bridges, rail transit and other infrastructure to improve urban transportation convenience and environmental quality. Second, we will strengthen the optimization and transformation of all official websites. Promote the construction of communication network, promote all types of network line network to work on the ground, clean the road environment. Third, we will strengthen environmental protection and improve the ecological quality. The government should increase investment in environmental protection, strengthen the construction of environmental protection facilities such as urban greening, sewage treatment and garbage classification, and relevant departments should form basic work plans and action plans to improve the quality of the urban environment and improve the living comfort of residents. Fourth, we will encourage green and low-carbon life and advocate energy conservation and emission reduction. The government should step up the promotion of low-carbon technologies such as green buildings and new-energy vehicles, encourage residents to adopt energy-saving and environmentally friendly lifestyles, and reduce energy consumption and environmental pollution. We will improve community services and strengthen community management. Fifth, we will promote the government's management level and improve its management level. We should strengthen the construction of community service facilities, improve the level of community management, and provide a convenient and comfortable living environment for the residents.



Figure 4-5: a schematic diagram of the road system optimization(source: modified by the author)



• Improve car charging facilities



• Shared bike distribution point

Figure 4-6: diagram of other improvement measures(source: Online pictures)

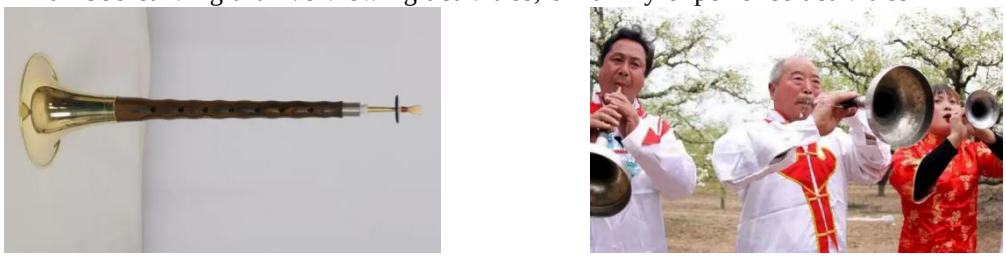
4.3.5 Conduct activity planning to form characteristic industries and explore the connotation of healthy towns

In addition to the construction of the material space part, excavating the spiritual and cultural deposits of small towns is an important part of the town. Dujiangyan city carries the responsibility of implementing the world famous tourist city, so it is quite necessary to carry out the characteristic construction of the surrounding small towns, especially the cultural planning of special seasons. As a world cultural heritage site, Dujiangyan city has a rich historical and cultural heritage and natural tourism resources. Dujiangyan city and its surrounding towns can fully tap these resources, create tourism projects with regional characteristics and cultural connotation, and then cultivate

the new power of the development of the towns. For example, a historical and cultural corridor can be set up in the town park to show the local history and culture, folk customs, folk art, etc. In the service of surrounding towns, local characteristic tourism projects should be added, and characteristic activities related to local culture should be added. Tianguan Festival, Yaowang Temple Association, Juyuan bamboo carving and Tianma sedan room suona can play their role at this time. In addition, these small towns around Dujiangyan city can also rely on the advantages of Dujiangyan water conservancy project to develop tourist attractions with water conservancy project as the core, such as water conservancy project exhibition hall, water conservancy project science education base, etc. Finally, we can promote the characteristic and healthy urban development path, create the theme of healthy development, combine the water system and health publicity, create a healthy cultural and tourism life, and scientifically integrate into the surrounding groups.



• Bamboo carving art live viewing activities, or family experience activities



• Open suona art activities and recreational activities for the elderly



• Health and sports activities publicity, physical examination coverage of the elderly, healthy diet activities casting in small town

Figure 4-7: Diagram of cultural activities(source: Online pictures)

4.3.6 Promote digital empowerment, strengthen coordination and cooperation, and implement digital urban and rural development

Promoting digital empowerment is the key to achieving efficient coordination between urban and rural construction. With the rapid development of science and technology, digital technology is penetrating into all aspects of our life. The construction of such key towns around Dujiangyan should also keep up with the pace of The Times and actively promote digital empowerment. First, we should promote the digitization of urban and rural infrastructure. By building a unified data platform, the data of various departments and levels are integrated to realize real-time data sharing. At the same time, big data, cloud computing and other technologies are used to conduct in-depth data mining and analysis, so as to provide accurate and scientific decision-making basis for urban and rural construction. Secondly, the digitization of urban and rural planning should be strengthened. With the help of GIS and In-SAR technology, the spatial data model is established to carry out a comprehensive planning for urban and rural construction. Through real-time monitoring of data changes, the planning is dynamically adjusted to ensure the scientificity and rationality of urban and rural construction. In addition, to promote the cooperation in digitization, in the past, the construction of digital platform needed capital and the technology was not mature. Now many enterprises have mature and open trial platforms, and in view of the capital problems and talent problems, it can promote the construction of industry-university-research platform, promote the healthy employment of students, and regain confidence in work. Moreover, the digitization of urban and rural industrial chains should be promoted. The Internet of things and artificial intelligence technologies will be used to promote the digital transformation of the industrial chain and improve the production efficiency. Through the Internet platform, the urban and rural

industries and the market are closely connected, realizing the real-time matching of supply and demand information, and injecting new vitality into the urban and rural economic development.

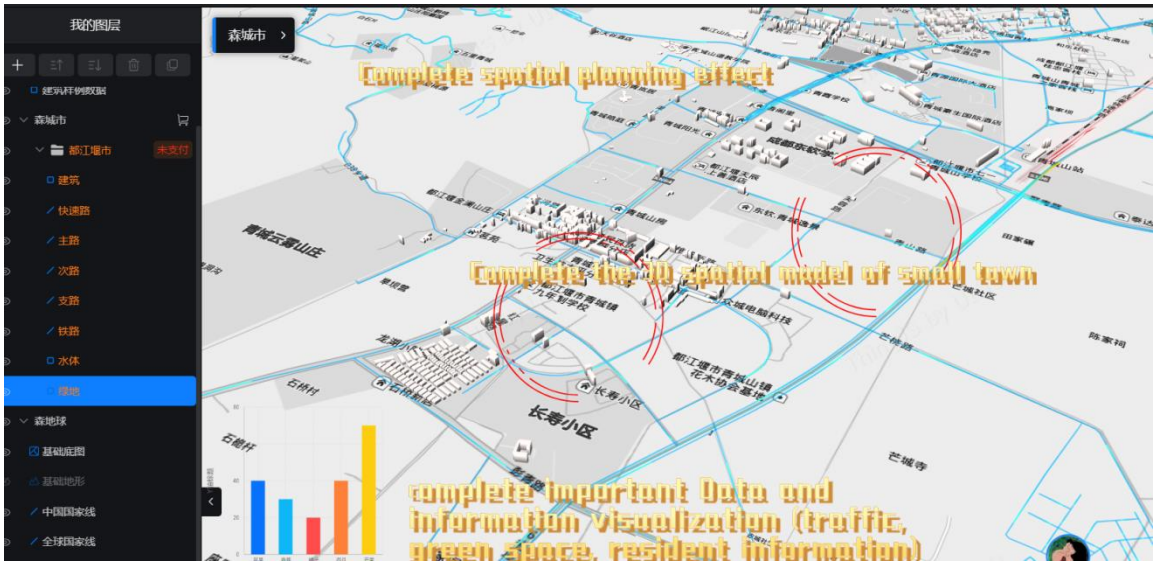


Figure 4-8: Diagram of digital management (source: modified by author)

5. OPTIMIZATION MEASURES OF SMALL TOWNS IN THE SUBURBS OF DUJIANGYAN CITY

Based on the analysis of the current situation of case small towns, this paper summarizes the existing problems of the current situation and the optimization points, and puts forward improvement measures, which has certain reference value for small towns with similar characteristics and similar development levels. At present, the proposed activity planning and management means only put forward ideas and strategies, and its specific implementation also needs to formulate detailed measures and plans. At the same time, for other small towns whose development level is significantly higher than or lower than the case small towns, they need to be adjusted according to the actual situation.

Due to the dynamic evolution of small towns, the renovation and improvement strategy proposed in this paper is based on the current situation, which needs to be adjusted according to the time and local conditions.

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